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SECTION:H

Household Carbon Footprint Target Tracker

Architectural Selection & Justification

1. Architecture Selection

We chose **Layered Architecture** for the Household Carbon Footprint Target Tracker system. The architecture is divided into three layers: Presentation Layer, Business Layer, and Data Layer. Each layer performs a specific responsibility and communicates with other layers.

2. Reasons for Choosing Layered Architecture

Reason 1: Separation of Responsibilities

Layered Architecture separates the user interface, carbon emission calculations, target tracking, and database operations into different layers. This makes the system easier to understand, develop, test, and maintain. Changes in one layer can be managed without unnecessarily affecting other layers.

Reason 2: Easy Maintenance and Expansion

The system can be expanded with additional features such as monthly emission reports, new emission categories, and progress monitoring. Separating the business logic from the user interface helps developers modify and maintain the application more easily.

3. Security Advantage

Layered Architecture allows security checks to be implemented in appropriate layers. The application can validate user input in the Presentation Layer and apply access control and data validation before storing information in the Data Layer. This helps reduce the risk of invalid or unauthorized data operations.

4. Performance Benefit

Separating calculation logic from the user interface allows carbon footprint calculations to be handled independently. The system can process activity data and calculate emissions without placing all processing responsibilities on the user interface. Efficient database queries and organized data processing can help improve system responsiveness.

Conclusion

Layered Architecture is suitable for the Household Carbon Footprint Target Tracker because it provides clear separation of responsibilities, easier maintenance, and structured communication between components. The architecture supports future feature development while organizing the system into manageable layers.

5. Component Diagram

